

BIOMECHANICS (EPHE 341)

Linear Kinetics

Impulse—momentum and friction · McGinnis
Chpt 3

Objective

To learn more kinetic quantities

- Impulse momentum relationship
- Friction

 Jumpers pushing against the ground

 A curling rock on ice

- From Newton's first 2 laws we learned that an external force is required to change the state of object motion (N1 and N2 [$F=ma$]) and an object in motion has momentum (N1).
- **But how does the momentum of an object change?**
- $F=ma$ (N2) which means that a force will cause an object to accelerate.
- Therefore force is proportional to acceleration
- change in velocity is the area under an acceleration time graph.
- The time that the object is accelerating and the magnitude of the acceleration will determine the velocity of the object

- If we want to consider change in momentum we need to multiply both sides by mass
- We know that $m\Delta v$ is momentum and $F = ma$
- Therefore

$$F = ma \rightarrow Ft = m\Delta v$$

- Impulse is a measure of a force applied for a certain amount of time
- Impulse = Force $\times$ time
- **$I = Ft$**
- The net force acting over some time will cause a change in momentum.
- impulse is the area under a force time graph

$$a = \Delta v / t$$

$$F = m(\Delta v / t)$$

$$Ft = m\Delta v$$

Impulse–momentum relationship

changes in the momentum of a system depend on:

1. the magnitude of external forces, and
2. the time over which the forces act

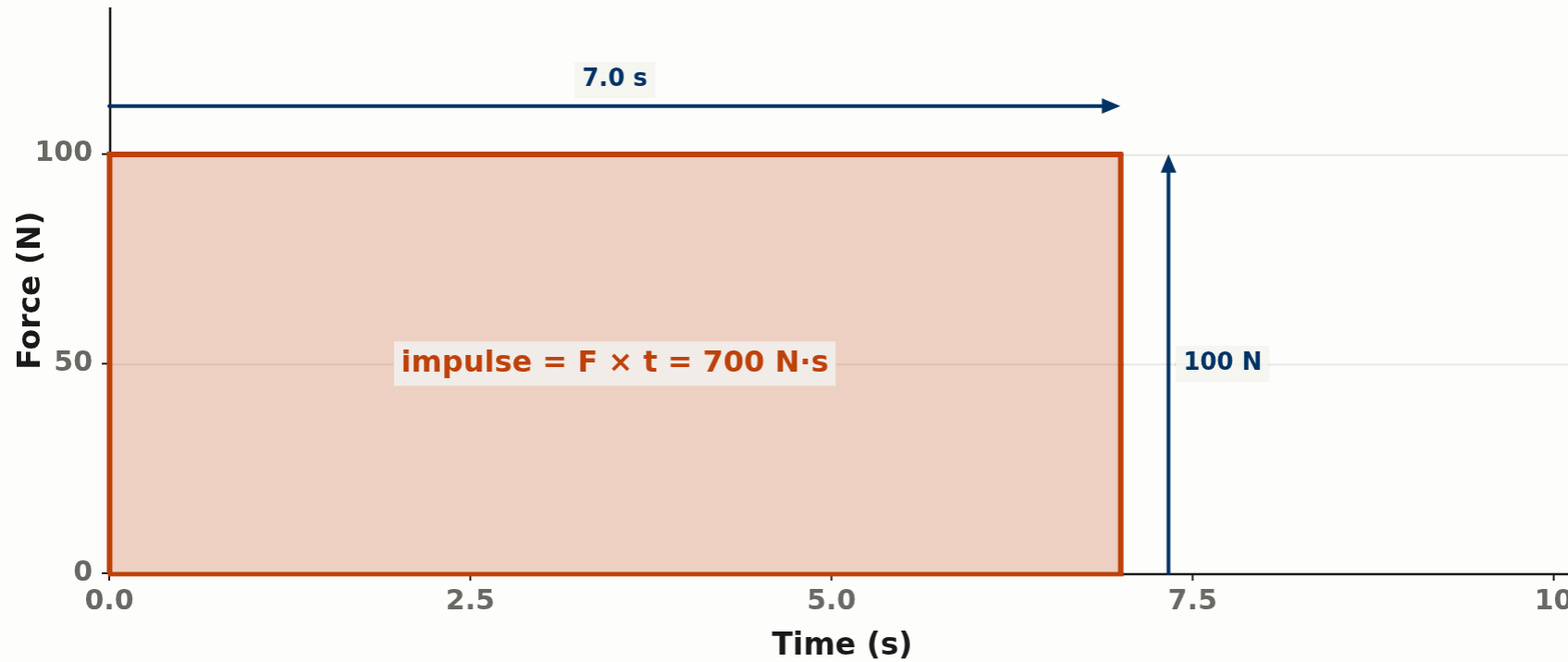
Therefore an impulse can cause a change in momentum

SI Units: N·s · Impulse is a vector

$$Ft = m\Delta v$$

$$Ft = (mv)_{\text{final}} - (mv)_{\text{initial}}$$

Impulse is an area



$$F \cdot t = 700 \text{ N} \cdot \text{s} \quad \Delta v = J / m = 700 / 90 = 7.78 \text{ m/s}$$

Impulse is the **area under a force–time graph**, and it is equal to the change in momentum. Halve the force and double the time and the area — and the change in velocity — is exactly the same. That is why a long push and a hard shove can do the same job, and why landing softly hurts less than landing stiffly.

Force Time

Mass

bobsled: 100 N for 7 s 50 N for 14 s 700 N for 1 s

Example Problem

During a bobsled race, two crew members push a 90 kg sled with an average force of 100 N over a period of 7 s before jumping in. Assuming that the sled started at rest, what is the bobsled's velocity after 7 seconds?

 A bobsled crew pushing at the start

Solution

given in this question:

- $F = 100 \text{ N}$
- $t = 7 \text{ s}$
- $m = 90 \text{ kg}$

$$Ft = (mv)_{\text{final}} - (mv)_{\text{initial}}$$

$$(100 \text{ N})(7 \text{ s}) = (90 \text{ kg})v - (90 \text{ kg})(0 \text{ m/s})$$

$$v = \mathbf{7.78 \text{ m/s}}$$

The bobsled's velocity is 7.78 m/s in the direction of force application

 A bobsled crew pushing at the start

$$\text{Impulse} = \text{Force} \times \text{time}$$
$$= \Delta \text{momentum}$$

- During human movement we are constantly trying to change the velocity of something.
- Walking (change the velocity of our center of gravity)
- Sports (change the velocity of our whole body or object)
- In these activities the force profiles can be explained by the impulse momentum relationship.



Ground reaction force profiles for walking and running

Neptune & McGowan, J Biomech 2011

Positions during a jump

Here are some basic positions during a jump that we will use to discuss kinetic and kinematic data during jumping

Linthorne, Am. J. Phys. 2001

 Body positions through a countermovement jump

 Vertical ground reaction force during a countermovement jump

Linthorne, Am. J. Phys. 2001

The jump: velocity

 Vertical velocity during the jump

 Vertical displacement during the jump

Linthorne, Am. J. Phys. 2001

Example Problem

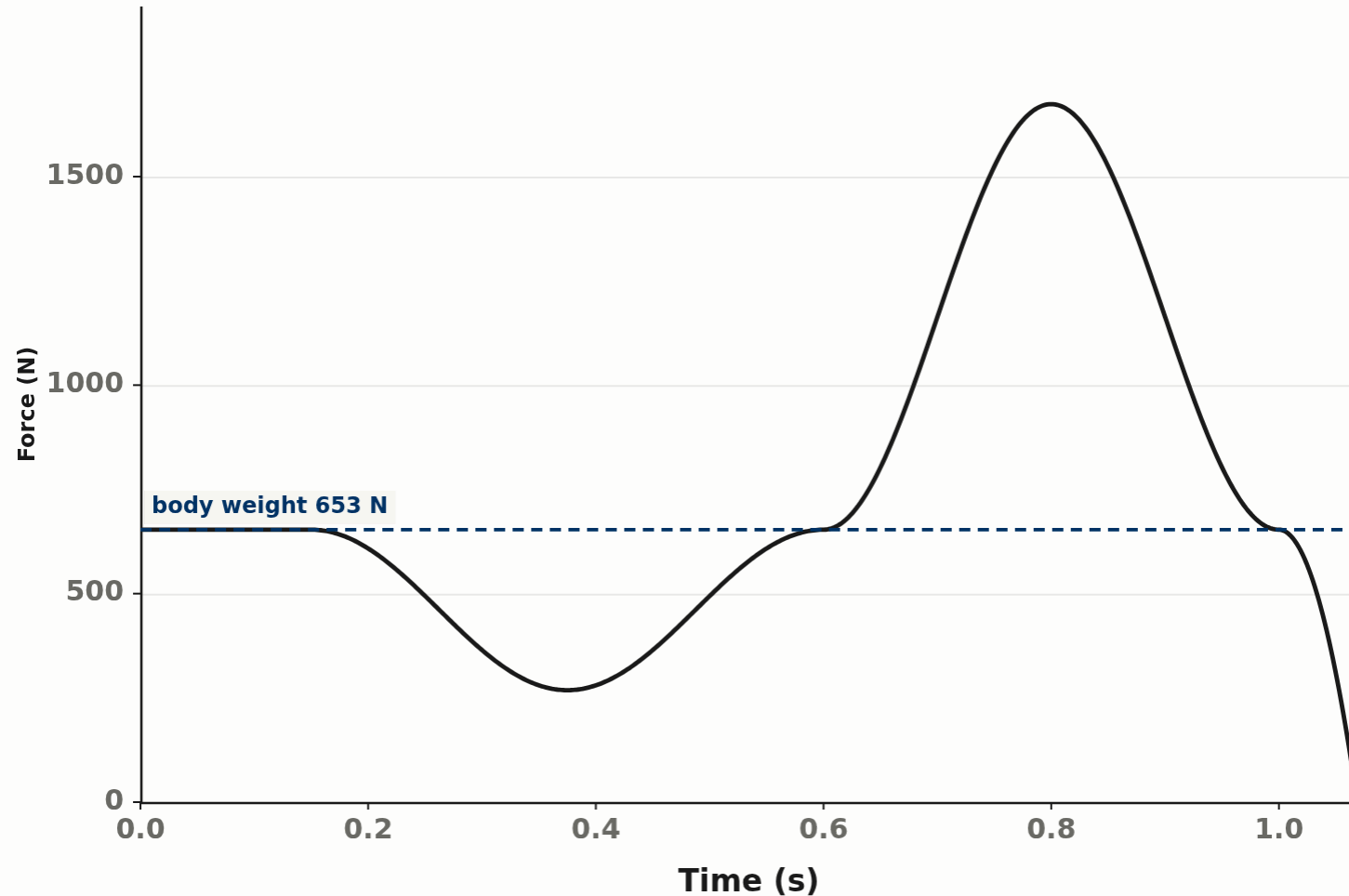
Using the force data of a jump from the following slides, If the jumper has a mass of 66.6 kg, what is his vertical velocity at the instant of take-off? What is the maximum height jumped?

 Positions through a jump

First: subtract body weight

the force trace

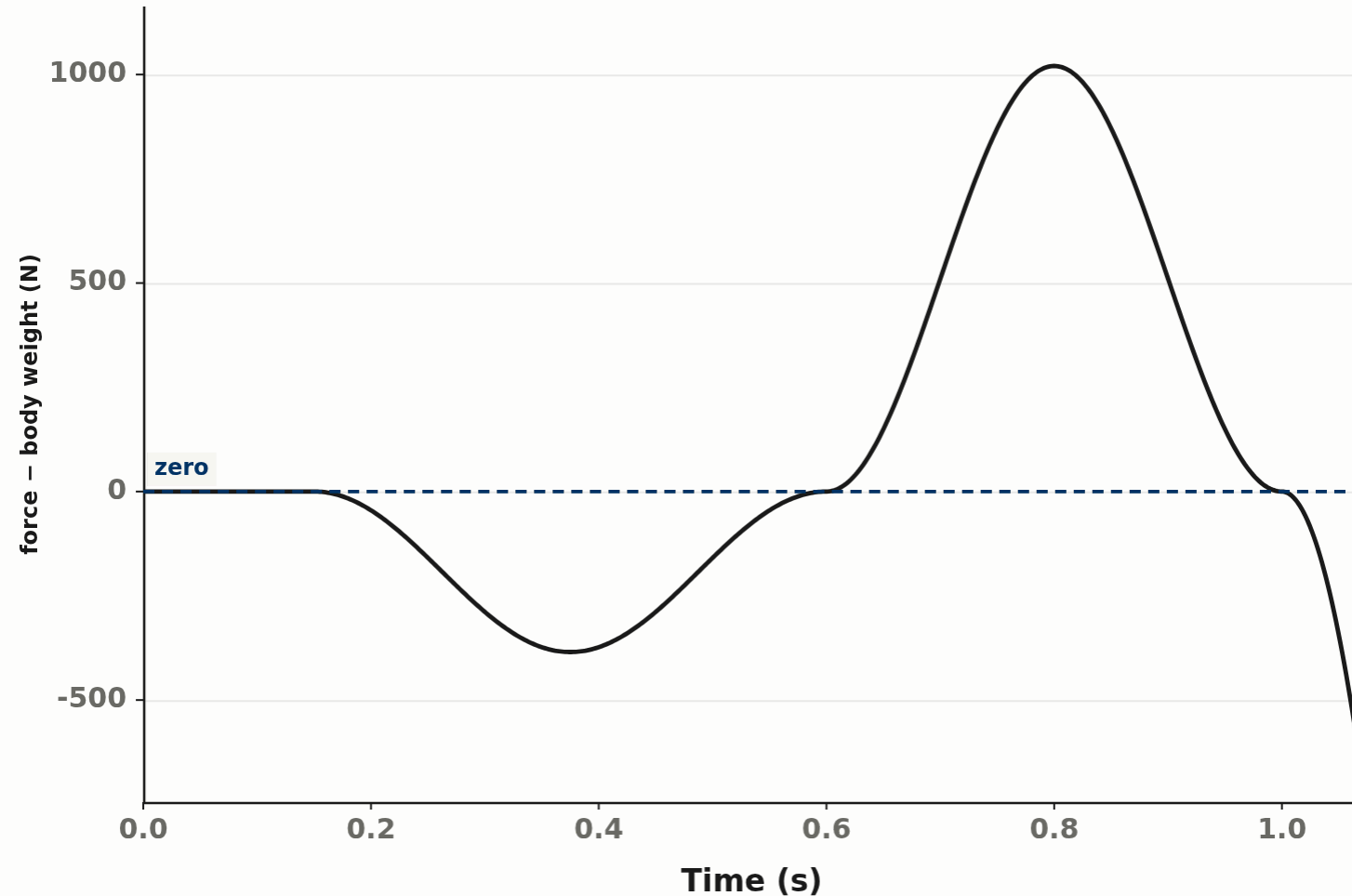
A countermovement jump measured on a force plate. Before it means anything we have to know where **body weight** sits on it — that is the dashed line, 653 N.



net impulse **102.5 N·s** · take-off **1.54 m/s** · height **12 cm**

The height is decided by the **net** impulse — the positive area minus both negative ones. That is why the answer is not just the big push: everything the jumper did before it is still on the books.

What is left is the net force



subtract body weight

First, subtract the jumper's body weight from the force vs time data. What is left is the **net** force, and its area is the change in momentum.

net impulse **102.5 N·s** · take-off **1.54 m/s** · height **12 cm**

The height is decided by the **net** impulse — the positive area minus both negative ones. That is why the answer is not just the big push: everything the jumper did before it is still on the books.

The three phases

1. $\text{Weight} > \text{ground reaction force} = \text{negative change in momentum}$
2. positive impulse which stops the downward motion and then, results in a positive change in momentum = vertical velocity that results in the jump.
3. negative impulse slows the upward motion of the jumper just prior to take-off.

The height the individual jumps is determined by the **net impulse**: That is, the difference between the total positive impulse and the total negative impulse.

Why add up all three?

- So why do we add up the 3 different impulses?
- Why not just the positive one?
- We need to account for all of the prior impulse to determine what will happen

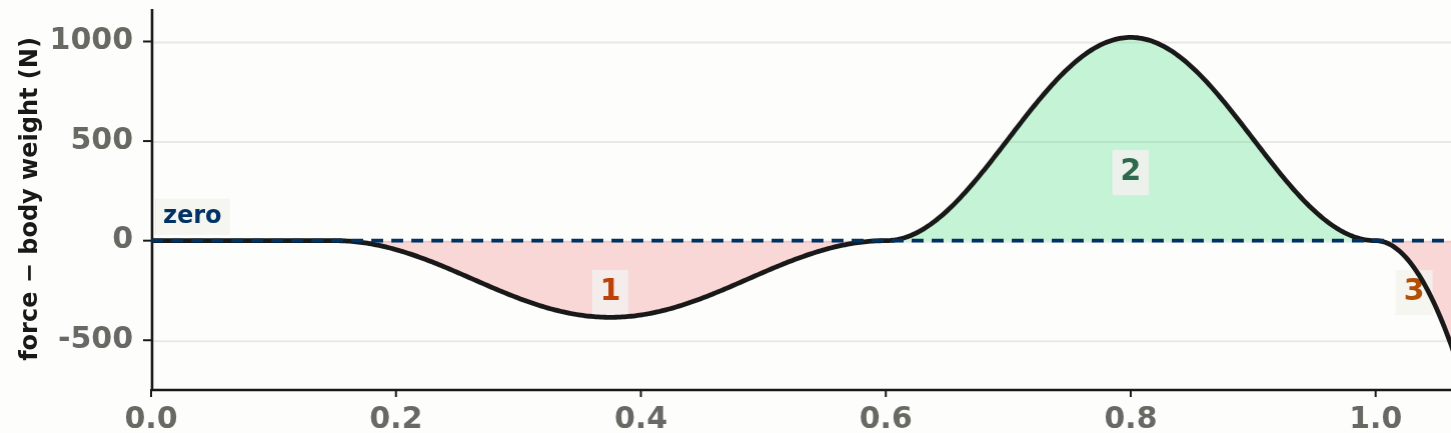
The first negative impulse has already given the jumper downward momentum. Part of the positive impulse is spent undoing it before any of it goes into going up.

Velocity through the jump

Linthorne, Am. J. Phys. 2001

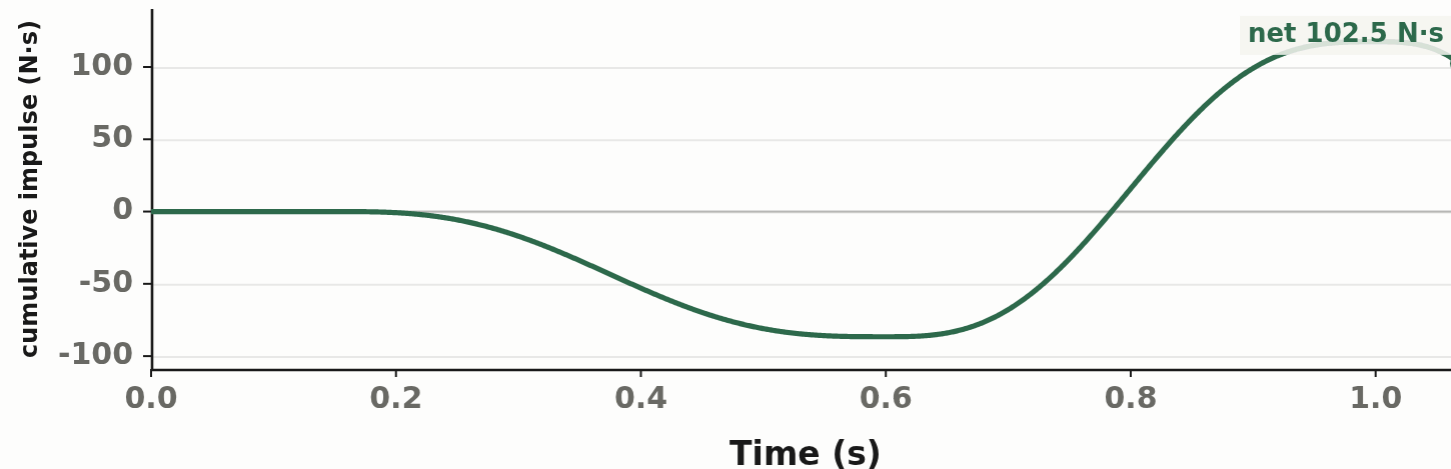
Next: take the area of each phase

Next, we want to calculate the area under the force vs time graph (the impulse) for all three phases of the jump. We do this by multiplying each force data point by the time interval between points and then summing all areas (remember integral sum)



the three impulses

	impulse
1st negative	-86.6 N·s
positive	204.1 N·s
2nd negative	-15.0 N·s
net impulse	102.5 N·s



The three impulses

1st Negative Impulse	-86.6 N·s
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Positive Impulse	204.1 N·s
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2nd Negative Impulse	-15.0 N·s
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Net Impulse	102.5 N·s
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 The jump force trace with its three phases

Example Problem (continued)

- since impulse is equal to the change in momentum, we can think of the jump in this way:

$$Ft = (mv)_2 - (mv)_1$$

- since the mass of the jumper doesn't change, we can factor it out to simplify the equation:

$$Ft = m(v_2 - v_1)$$

Example Problem (continued)

we know that the jumper begins at rest ($v_1 = 0$); therefore,

$$Ft = m(v_2 - 0)$$

$$Ft = mv_2$$

$$102.5 \text{ N}\cdot\text{s} = (66.6 \text{ kg}) v_2$$

$$v_2 = \mathbf{1.54 \text{ m/s}}$$

where v_2 is the velocity the instant before take-off.

The 102.5 N·s is the **net impulse**.

 Velocity reaching its peak at take-off

Example Problem (continued)

to calculate the maximum height jumped, we have to use one of the Equations of Constant Acceleration (we will cover this in projectile motion)

$$v_y(t) = -9.81t + v_{oy}$$

at the maximum height, the jumper will be at rest ($v_f = 0$)

$$0 = -9.81t + 1.54 \rightarrow t = 0.157$$

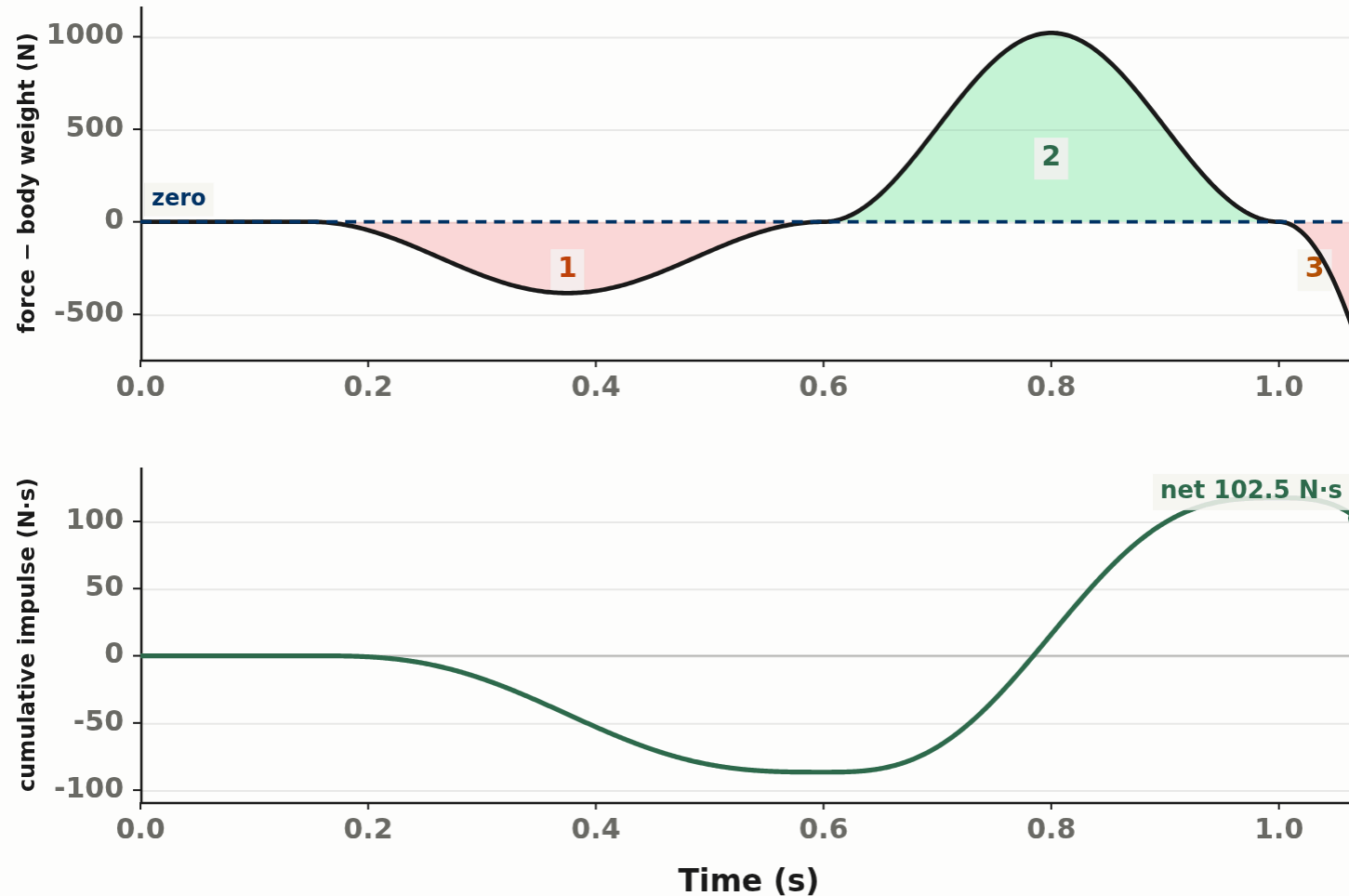
$$d_y(t) = (-9.81/2)t^2 + v_{oy}t + d_{oy}$$

$$d_y(t) = (-4.905)(0.157)^2 + (1.54)(0.157) + 0$$

$$d_y(t) = -0.12 + 0.24$$

 Displacement through the jump and flight

The whole calculation



velocity and height

	impulse
1st negative	-86.6 N·s
positive	204.1 N·s
2nd negative	-15.0 N·s
net impulse	102.5 N·s

$$Ft = m(v_2 - v_1), \text{ and } v_1 = 0$$

$$102.5 = (66.6) v_2$$

$$v_2 = 1.54 \text{ m/s}$$

rise to the peak

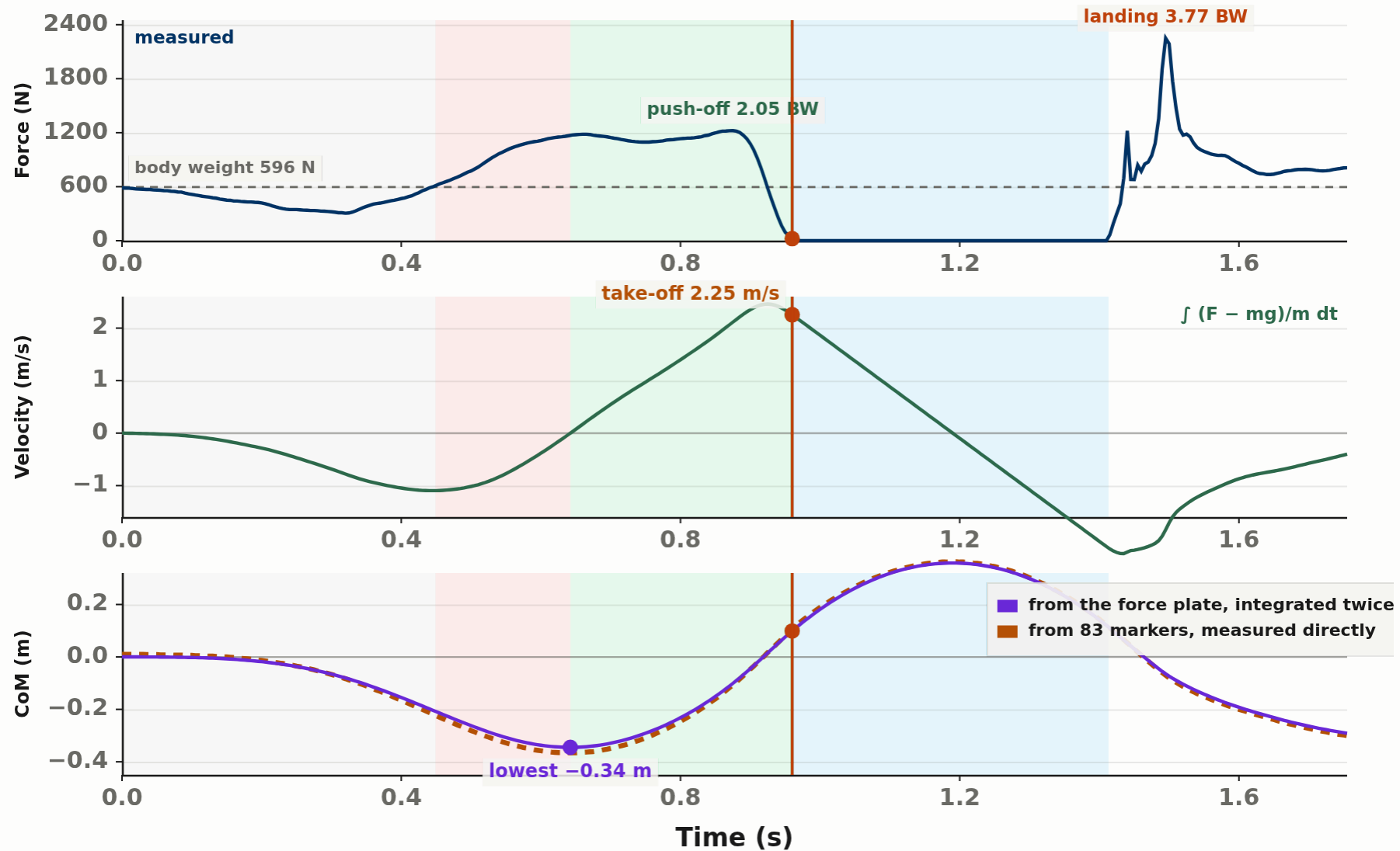
$$h = v^2 / 2g = 0.12 \text{ m}$$

net impulse **102.5 N·s** · take-off **1.54 m/s** · height **12 cm**

The height is decided by the **net** impulse — the positive area minus both negative ones. That is why the answer is not just the big push: everything the jumper did before it is still on the books.

Integrate it twice

A measured jump — force, then velocity, then the centre of mass



How high did he jump?

Two answers, from two different parts of the record

FROM THE FORCE PLATE

impulse → velocity → height

$$\int (F - mg) dt = m \cdot v$$

$$v = 2.255 \text{ m/s}$$

$$h = v^2 / 2g$$

$$h = 2.255^2 / (2 \times 9.81)$$

25.9 cm

FROM THE FLIGHT TIME

the plate reads zero for 0.453 s

$$t_{\text{air}} = 0.453 \text{ s}$$

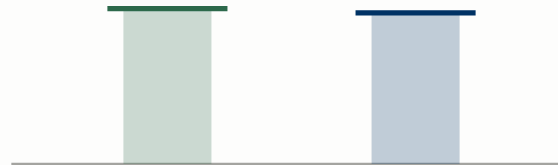
up and down are equal,

$$h = g \cdot t_{\text{air}}^2 / 8$$

$$h = 9.81 \times 0.453^2 / 8$$

25.2 cm

they differ by 7 mm (2.8%)

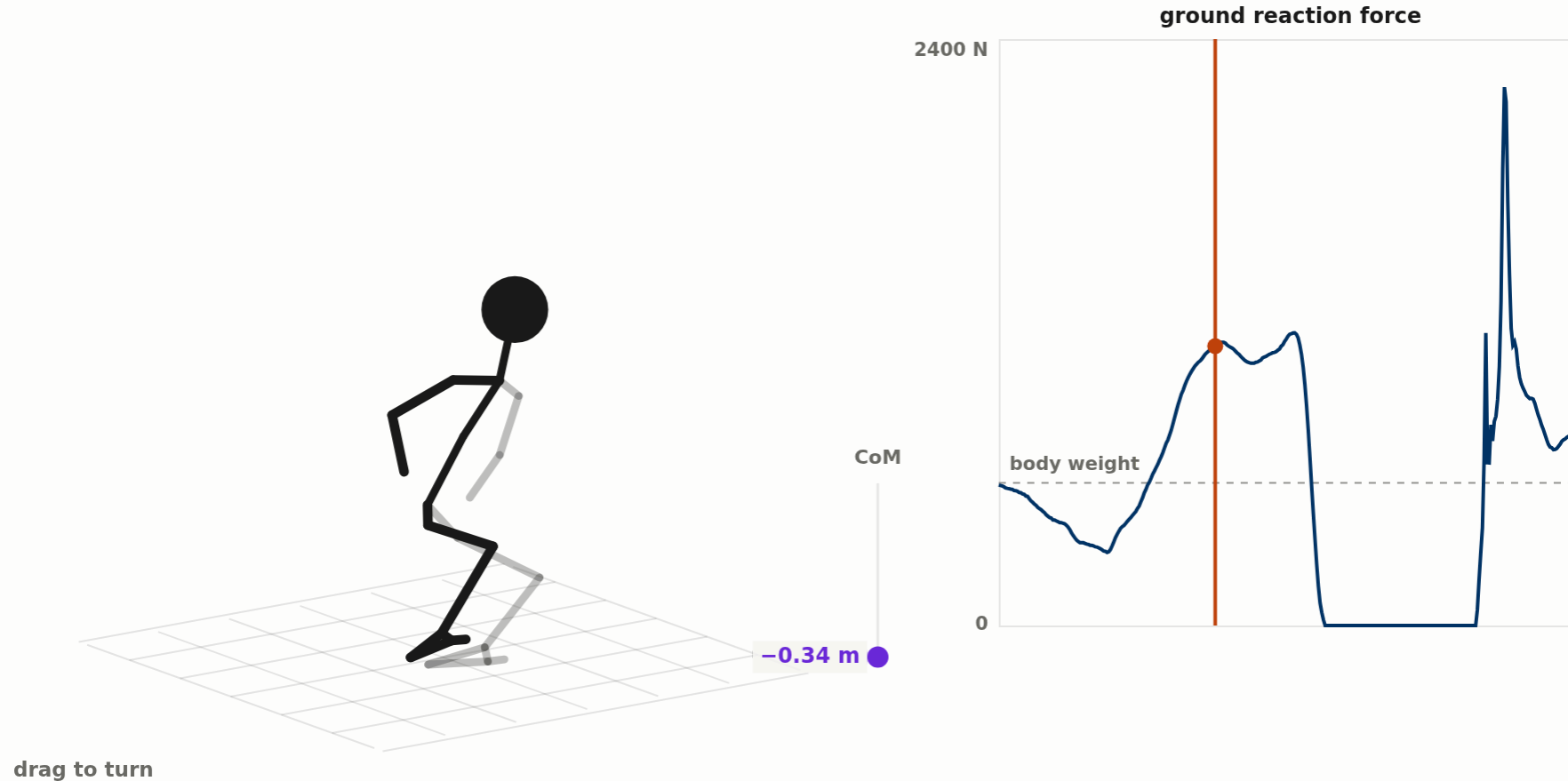


Two measurements, sharing nothing but the same jump: **25.9 cm** and **25.2 cm** · the force route never uses the flight time, and the flight-time route never uses the force

The same jump, in three dimensions

Twenty-two measured joint centres — drag to turn it

countermovement — eccentric



F 1167 N · v -0.02 m/s · CoM -0.34 m · SUSU-46, 83 markers at 90 Hz and two force plates at 450 Hz, on one clock

Understanding impulse/momentum and how it relates to the force time graph gives us the ability to interpret activity from the force profiles.

 Force record of ten activities in sequence

Headon and Curwen 2001 · Stand, Sit, Drop-landing, Jump, Crouch, Rise to stand, partial weight (feet only), unloaded

 Legend for the activity record

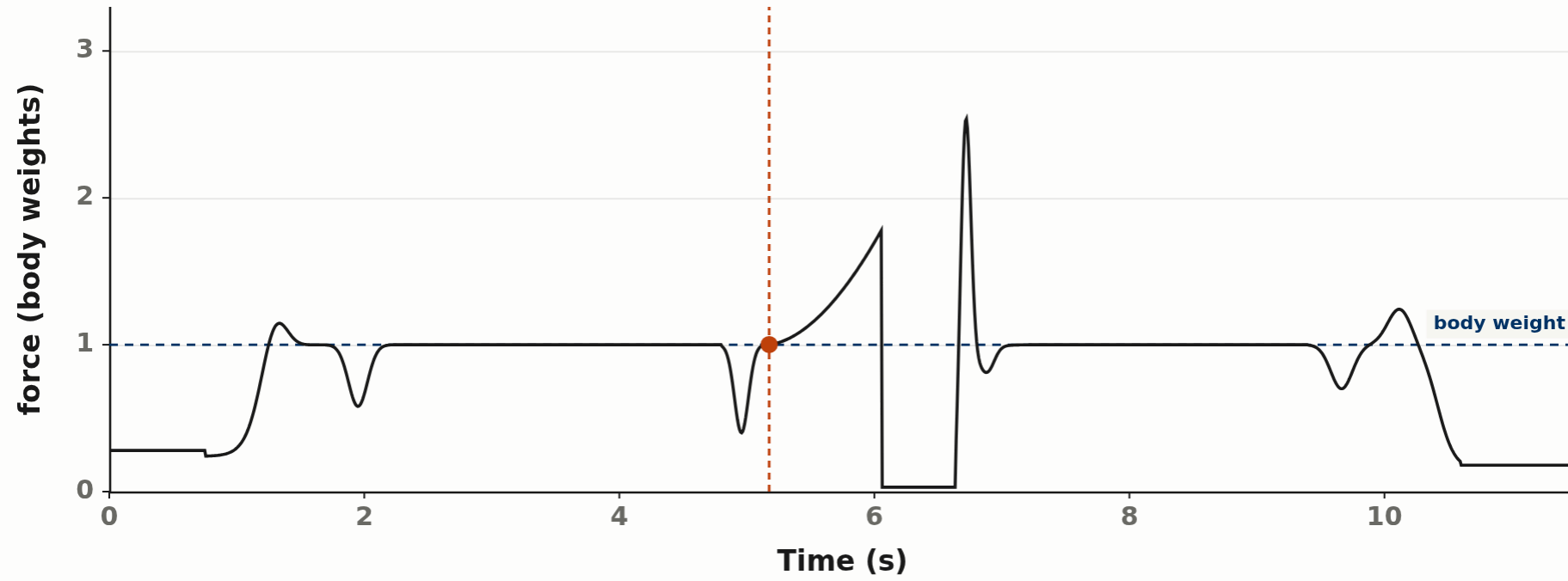
Reading activity off a force profile



A second activity record

Headon and Curwen 2001 · Stand, Sit, Drop-landing, Jump, Crouch, Rise to stand, partial weight (feet only), unloaded

Name the activity



$$t = 5.2 \text{ s} \cdot 1.00 \text{ BW} = 737 \text{ N}$$

Ten things done in a row on one force plate. Before revealing the labels, work out from the trace alone which is the jump, which is the drop landing, and where the person is sitting down — the shapes are the argument.

Cursor

Reveal the activities

5.2 s

Friction: The force resisting the relative motion of solid surfaces.

- Results from the interaction of the surface area of objects in contact
- **Dry Friction** — Static, kinetic
- Fluid
- Lubricated
- Skin
- Internal

 Two surfaces in contact magnified

Friction is important when considering interaction and mobility of the whole body with the surroundings as well as proper functioning of internal anatomical structures.

Tribology – the science and engineering of interacting surfaces in relative motion

-  A climber relying on friction
-  Running shoe outsoles
-  A knee joint

- a force that acts at the surface of contact between two bodies opposite in the direction of motion or impending motion
- consider a crate sitting on the ground:
- when a small force is applied to one side of the crate, it remains motionless
- the force that opposes the applied force pushing the box forward is the force of static friction

$$F_{\text{applied}} \longleftrightarrow F_{\text{sfr}}$$

- as F_{applied} increases, F_{sfr} increases until a critical point is reached
- at the critical point, F_{sfr} is called the force of **maximum static friction** (F_{msfr})
- if F_{applied} increases further, the box will begin to slide

$$F_{\text{msfr}} = \mu_s N$$

where:

- μ_s is the coefficient of static friction
- N is the force that is normal (perpendicular) to the surface of contact
- F_{msfr} is the force of maximum static friction

the coefficient of friction indicates the ease of sliding — the greater μ , the less easily one surface will slide on another

 A boot sole on a surface

- the normal force (N) is equal but opposite the component of the object's weight acting perpendicular to the surface on which the object is sitting
- The force of friction acts parallel to the surface

$$mg = -N$$

APPROXIMATE COEFFICIENTS OF STATIC FRICTION (DRY)

steel & steel	0.8
glass & glass	0.94
concrete & rubber	1.0
steel & ice	0.005

 Normal force and friction force on a surface

- once the box is in motion the frictional force acting on the box is called kinetic friction (F_{kfr})
- as F_{applied} continues to increase, F_{kfr} remains constant

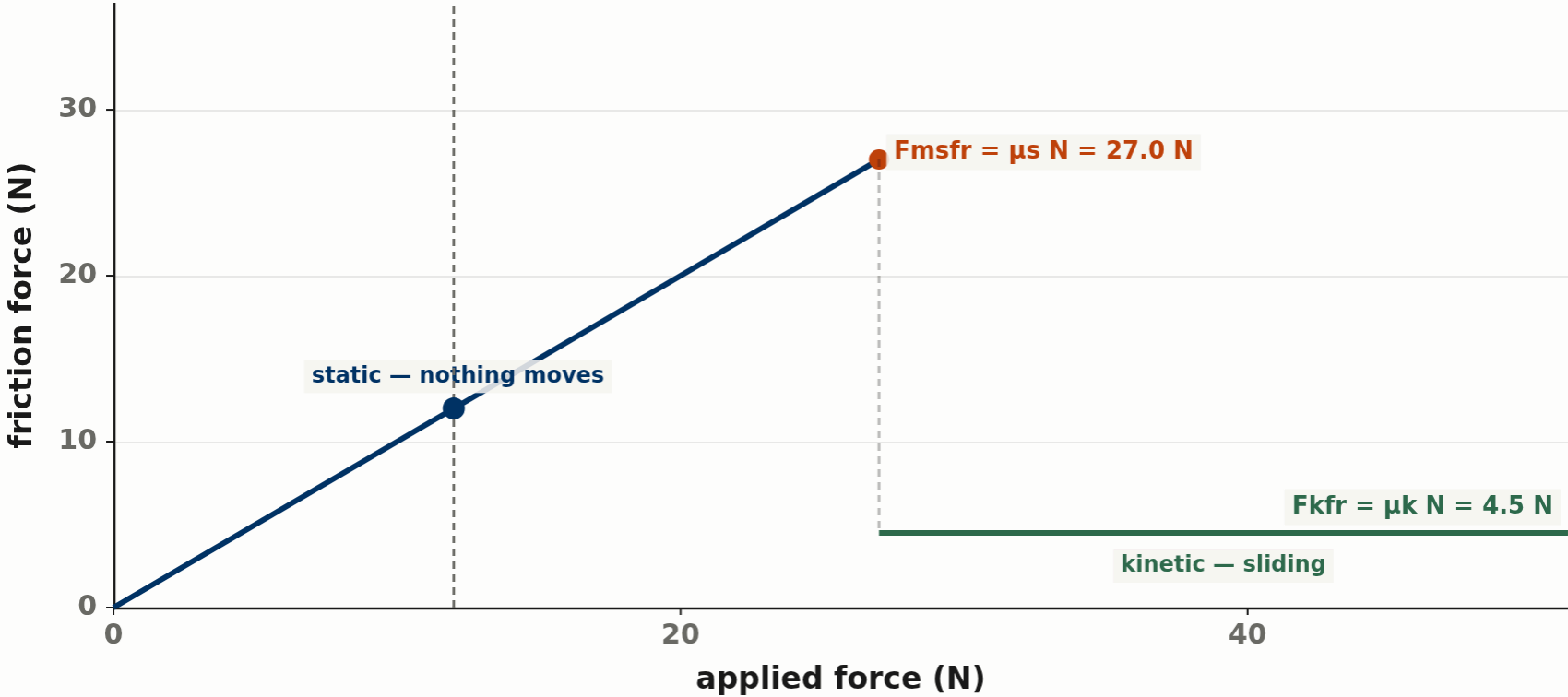
$$F_{\text{kfr}} = \mu_k N$$

where:

- μ_k is the coefficient of kinetic friction
- N is the force that is normal (perpendicular) to the surface of contact
- F_{kfr} is the force of kinetic friction

 A curling rock sliding on ice

Static and kinetic friction together



pushing with **12.0 N** · friction pushes back with **12.0 N** · **still stuck**

Static friction is not a fixed number — it is whatever it needs to be, up to $\mu_s N$. Push with 5 N and it pushes back with 5 N. The coefficient indicates the ease of sliding: the greater μ , the less easily one surface slides on another.

curling rock: 90 N, 0.3 / 0.05 rubber on concrete: 1.0 steel on ice: 0.005

Normal force

90 N

Applied force

12.0 N

μ static

0.300

μ kinetic

0.050

Example Problem

The coefficient of static friction between a 90 N curling rock and the ice is 0.3 and the coefficient of kinetic friction is 0.05. How much force is required to start the rock in motion? How much force is required to keep the rock in motion?

 A curler delivering a rock

Solution

in this question, we are given: $\mu_s = 0.3$, $\mu_k = 0.05$, $N = 90 \text{ N}$

first, calculate the force of maximum static friction:

$$F_{\text{msfr}} = \mu_s N = (0.3)(90 \text{ N}) = \mathbf{27 \text{ N}}$$

next, calculate the force of kinetic friction:

$$F_{\text{kfr}} = \mu_k N = (0.05)(90 \text{ N}) = \mathbf{4.5 \text{ N}}$$

Just greater than 27 N are required to start the rock in motion and 4.5 N are required to keep it in motion.

 A curling rock

- Remember from our forces lecture, we calculated the perpendicular, and parallel components of gravity on a slant surface
- The parallel component of gravity will tend to pull the block down the surface.
- Friction force is required to be equal and opposite of the parallel force or the block would slide down the slope
- Friction force is proportional to the normal force and acts perpendicular to it.

 Gravity, normal and friction forces on a slope

Components on any incline

For any incline of a certain angle the parallel and perpendicular components of force will be

Perpendicular Force $F_{pd} =$
 $mg \cdot (\cos \theta)$

Normal Force $N = -mg \cdot (\cos \theta)$

Parallel Force $F_{pl} = mg \cdot (\sin \theta)$

Friction Force $F_{fr} = -mg \cdot (\sin \theta)$

The perpendicular axis is the **compression axis**; the parallel axis is the **shear axis**.

Friction and the normal force are the reactions: each one is equal and opposite to the component of gravity along its own axis.

Worked example: 4 kg at 20°

How to determine the force of friction on a slant surface. We need to know the angle of the slant (20°) and the mass of the object (4 kg).

1. Draw a diagram — the axis system, and all known and to be determined forces
2. Solve for the magnitude of the force of gravity
$$W = mg = 4 \text{ kg} \times -9.81 \text{ m/s}^2 = \mathbf{-39.24 \text{ N}}$$
3. Solve for the direction of the force of gravity relative to the axis system
4. Solve for the parallel and perpendicular components of force

 Free body diagram on a 20 degree slope

Worked example: the components

4th — Solve for the parallel and perpendicular components of force

Parallel component

$$\begin{aligned}\text{Opposite} &= H \cdot \sin 20^\circ = \\ -39.24 \cdot \sin 20^\circ &= \mathbf{-13.42 \text{ N}}\end{aligned}$$

Perpendicular component

$$\begin{aligned}\text{adjacent} &= H \cdot \cos 20^\circ = \\ -39.24 \cdot \cos 20^\circ &= \mathbf{-36.87 \text{ N}}\end{aligned}$$

5th — Solve for the Normal and friction forces

$$\text{Parallel component} = 13.42 \text{ N}$$

$$\text{Friction force} = -F_{pl} = \mathbf{-13.42 \text{ N}}$$

$$\text{Perpendicular component} = -36.87 \text{ N}$$

$$\text{Normal force} = -F_{pd} = \mathbf{36.87 \text{ N}}$$

Example Problem

An 80 kg hiker is standing on a sloped trail at an angle of 35° above horizontal. If the coefficient of static friction between the hiker's boot and the trail is 0.83, does the hiker slip?

 A hiking boot on a sloped trail

① *Draw a free-body diagram*

 Free-body diagram of a boot on a 35 degree slope

② *Calculate the weight of the object, perpendicular and parallel forces of the object on the slope*

 Free-body diagram of a boot on a 35 degree slope

$$F_w = mg$$

$$F_w = (80 \text{ kg})(-9.81 \text{ m/s}^2)$$

$$F_w = \mathbf{-784.8 \text{ N}}$$

$$F_{pd} = W (\cos 35^\circ)$$

$$F_{pd} = \mathbf{-642.9 \text{ N}}$$

$$F_{pl} = W (\sin 35^\circ)$$

$$F_{pl} = \mathbf{450.1 \text{ N}}$$

③ Calculate the Normal and friction forces

 Free-body diagram of a boot on a 35 degree slope

$$N = -F_{pd}$$

$$N = -(-642.9 \text{ N})$$

$$N = \mathbf{642.9 \text{ N}}$$

$$F_{fr} = -F_{pl}$$

$$F_{fr} = -(450.1 \text{ N})$$

$$F_{fr} = \mathbf{-450.1 \text{ N}}$$

This is the magnitude of frictional force required to maintain the boot on the slope

④ *Calculate the maximum force of static friction*

 Free-body diagram of a boot on a 35 degree slope

$$F_{\text{msfr}} = \mu_s N$$

$$F_{\text{msfr}} = (0.83)(642.9 \text{ N})$$

$$F_{\text{msfr}} = \mathbf{-533.607 \text{ N}}$$

⑤ *Determine if the maximum force of static friction is greater than the friction force*

 Free-body diagram of a boot on a 35 degree slope

$$F_{\text{fr}} = -450.1 \text{ N}$$

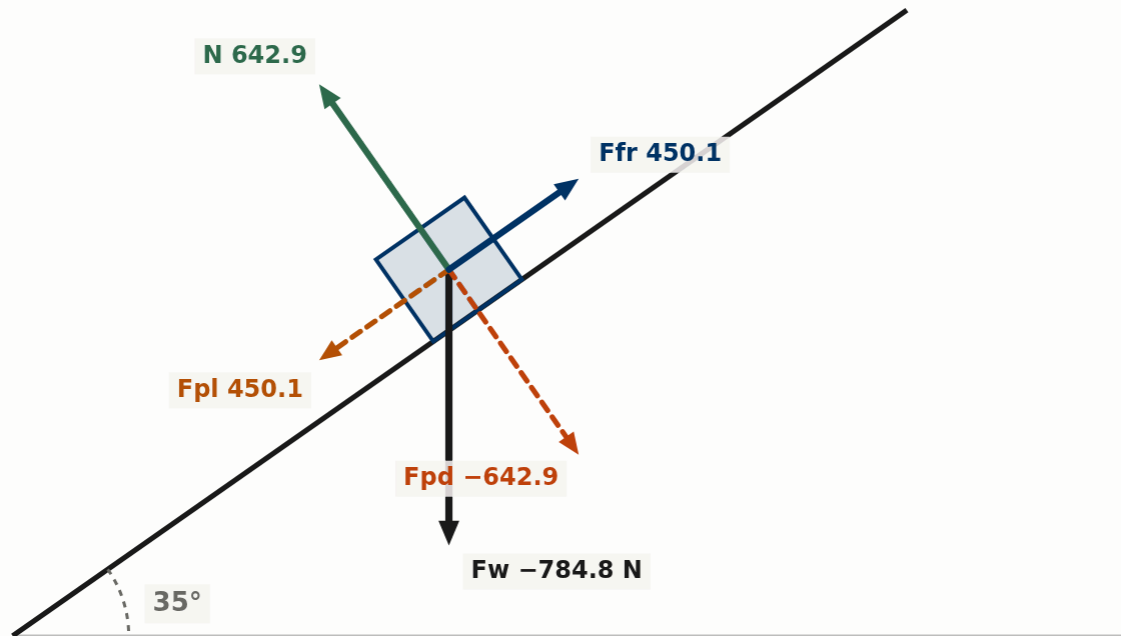
$$F_{\text{msfr}} = -533.6 \text{ N}$$

$$F_{\text{fr}} < F_{\text{msfr}}$$

Therefore, the hiker will not slip down the trail.

Friction

The same comparison, on a slope you can tilt



80 kg on a 35° slope, $\mu_s = 0.83$

$$F_w = mg = -784.8 \text{ N}$$

$$F_{pd} = W \cos 35^\circ = -642.9 \text{ N}$$

$$F_{pl} = W \sin 35^\circ = 450.1 \text{ N}$$

what the ground gives back

$$N = -F_{pd} = 642.9 \text{ N}$$

$$F_{fr} \text{ needed} = 450.1 \text{ N}$$

what the boot can supply

$$F_{msfr} = \mu_s N = 533.6 \text{ N}$$

needed
450

available
534

slips at
39.7°

$F_{fr} = 450.1 \text{ N} < F_{msfr} = 533.6 \text{ N}$ · **the hiker does not slip**

Both sides of the comparison move together as the slope steepens — the shear grows while the normal force, and therefore the friction available, shrinks. They cross at **39.7°**, where $\tan \theta = \mu_s$. Mass cancels out of that entirely: a heavier hiker is not any safer.

Investigation of the Frictional Response of Osteoarthritic Human Tibiofemoral Joints and the Potential Beneficial Tribological Effect of Healthy Synovial Fluid

Caligaris et al. Osteoarthritis Cartilage. 2009
October; 17(10): 1327–1332.

This study tests the hypothesis that the natural progression of osteoarthritis (OA) in human joints leads to an increase in the friction coefficient and that healthy synovial fluid (SF) may help mitigate the increase in the friction coefficient in diseased joints.

 A tibiofemoral joint

The friction coefficient of human tibiofemoral joints with varying degrees of OA was measured in healthy bovine SF and physiological buffered saline (PBS).

No statistical differences were found in the friction coefficient with increasing OA; however, the friction coefficient was significantly lower in SF than PBS in both configurations.



Friction coefficients measured in synovial fluid and saline

Caligaris et al., Osteoarthritis Cartilage 2009

- **Impulse** = Force $\times$ time = the area under a force–time graph
- $Ft = m\Delta v$ — an impulse causes a change in momentum. SI units N·s, and it is a vector
- Jump height is decided by the **net** impulse, not the positive one
- **Static friction** matches the applied force up to $F_{msfr} = \mu_s N$
- **Kinetic friction** is constant at $F_{kfr} = \mu_k N$
- On a slope, friction must supply $mg \cdot \sin\theta$ while the normal force falls to $mg \cdot \cos\theta$ — sliding begins where $\tan\theta = \mu_s$



Jumping



A bat striking a ball



A bicycle kick